

23CSE111 OBJECT ORIENTED PROGRAMMING

S2 B.Tech CSE

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – I & PHASE – II

Analysis and Architectural Modelling (CO1 & CO2)

Group No : _____ B8 _____

Team Members :

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PHASE – I (Analysis)

Problem Identification & Class Mapping (CO1)

1. Problem Title : _____ ATM Simulation System _____

2. Problem Description :An ATM allows customers to withdraw cash,deposit,and check account balance.

3. Identified Entities (Classes)

Class Name	Description
Customer	It stores the customer information.
Account	It stores the account details and the available balance.
Transaction	It stores the transaction details and type of transaction.
ATM	It handles ATM operations.

File Manager	It handles all the file operations.
DespositTransaction	It is used to add money in to the account
WithdrawTransaction	It is used to remove amount from the account safely.

4. Class Attributes and Behaviours

Class Name	Attributes	Methods (Behaviours)
Customer	name: String phoneNumber: int customerId: String account: Account	Customer(name: String, phoneNumber: int, customerId: String, account: Account) getAccount(): Account
Account	bankName: String accountNumber: String balance: double pin: int ArrayList<Transaction> transactions	Account(bankName: String, accountNumber: String, balance: double, pin: int) performTransaction(transaction: Transaction): void validatePin(enteredPin: int): Boolean getBalance(): double setBalance(balance: double): void showTransaction(): void
Transaction (Abstract)	transactionId: String, amount: double	Transaction(transactionId: int, amount: double) execute(account: Account): void toString(): String

ATM	isAuthenticated: Boolean fileManager: FileManager	Start(customer: Customer): void
DespositTransaction		DespositTransaction(transactionIdP: String, amount: double) execute(account: Account); void toString(): String
WithdrawTransaction		WithdrawTransaction(transactionId: String, amount: double) execute(account: Account): void toString(): String
FileManager		saveData(data: String): void readData(): void

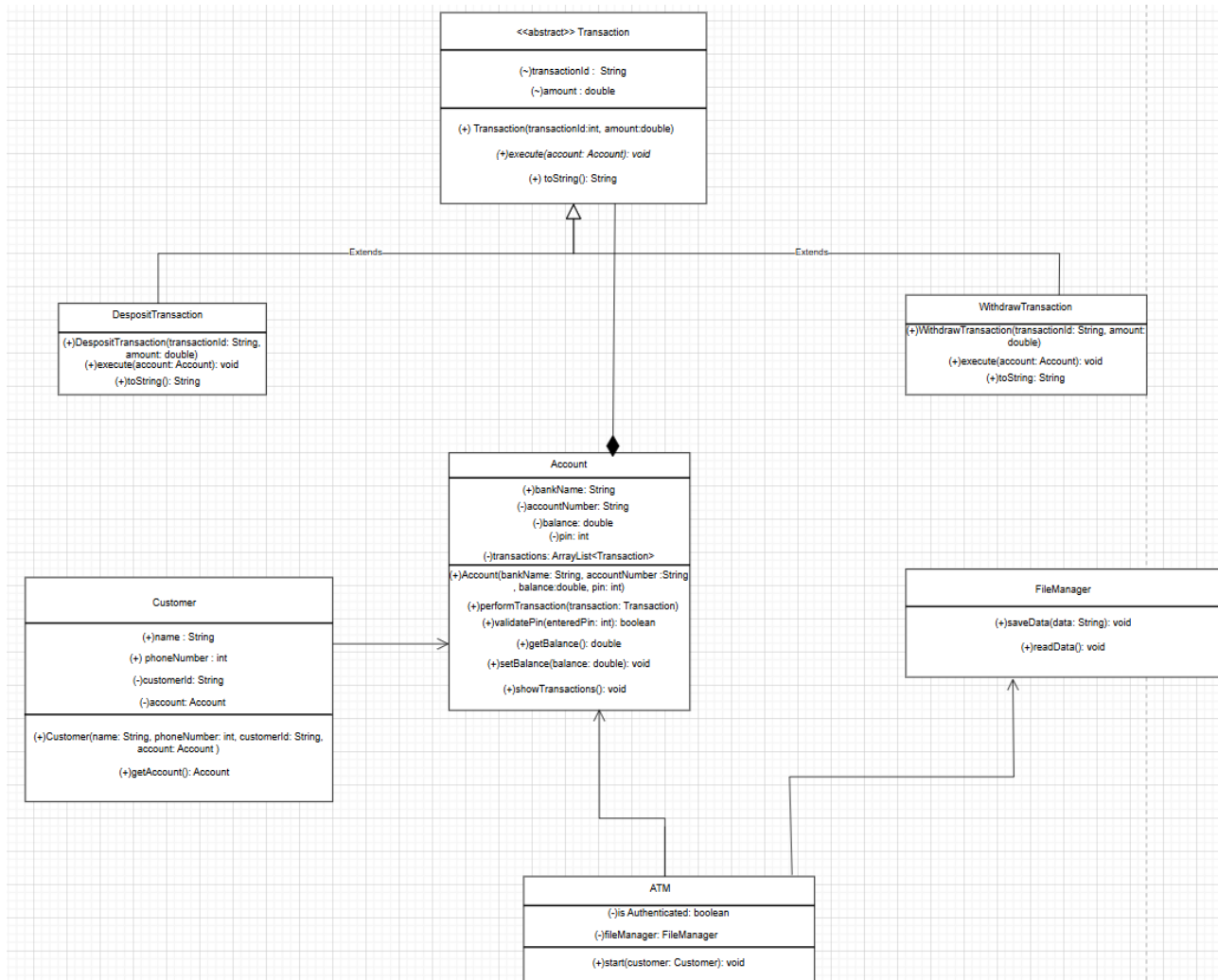
PHASE – II (Architectural Modelling)

UML Class Diagram and Sequence Diagram (CO2)

1. Problem Title : _____ ATM Simulation System _____

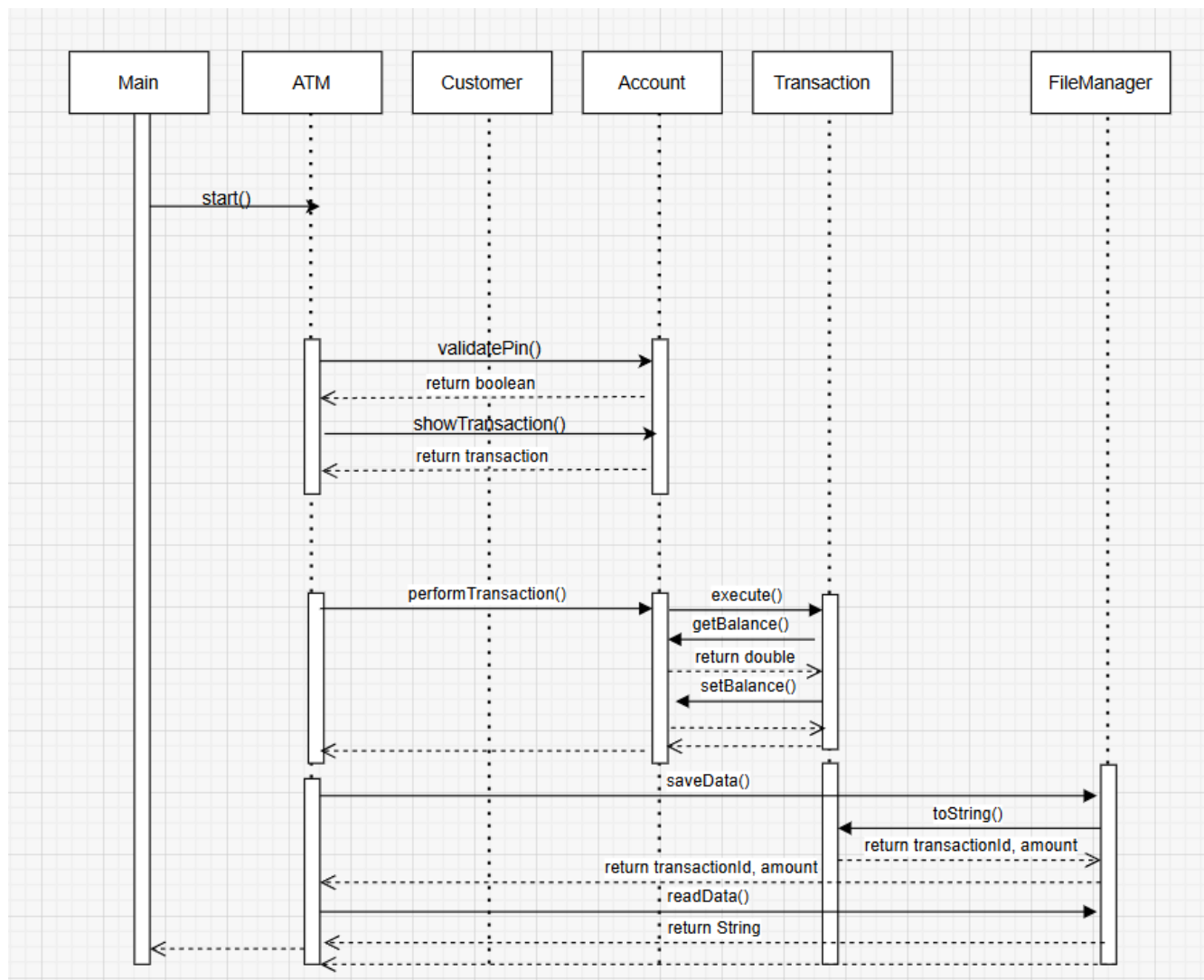
2. UML Class Diagram

Paste / Draw the UML Class Diagram here



3. Sequence Diagram

Paste / Draw the Sequence Diagram here



4. Brief Explanation of the Design

- Major classes identified and their responsibilities:
- Relationships used (association / aggregation / inheritance, if any):
- Key interaction flow captured in the sequence diagram:
 - 1) Customer
Stores basic details of the user and connects the user to their bank account.
 - 2) Account
Stores account number, balance, and PIN.
Also keeps a record of all transactions and performs deposit, withdraw operations.
 - 3) Transaction (Abstract Class)
A general class that defines what a transaction should have (ID and amount).
Used as a base for different types of transactions.
 - 4) DepositTransaction & WithdrawTransaction
These are some types of transactions. Deposit adds money, and Withdraw removes money from account.
 - 5) ATM
Checks the Pin and allows the user to choose options like deposit, withdraw, check balance, history.
 - 6) FileManager
Used to save and read data from a file (like storing transaction).

Relationships used:

- 1) Customer → Account (Association):
Used because a customer needs an account to access money, but both are separate classes.
- 2) ATM → Account (Association):
Used because the ATM needs the account to perform operations like deposit and withdraw.
- 3) Account → Transaction (Composition):
Used because all transactions are part of an account and created only when the account is used.
- 4) Transaction → Deposit & Withdraw (Inheritance):
Used because deposit and withdraw are different types of transactions that do the same thing in different ways.

Interaction Flow:

- 1) Main starts the ATM
- 2) ATM checks PIN with Account
- 3) If PIN is correct
- 4) ATM shows transaction details
- 5) ATM performs transaction like deposit, withdraw
- 6) Transaction updates balance
- 7) Account saves data using FileManager
- 8) ATM shows result to user